

- Q.27 Draw noise triangle, what is its significance? (CO3)
- Q.28 What is need of Pre-emphasis & De-emphasis in F.M.? (CO3)
- Q.29 Write any four advantages of FM. (CO4)
- Q.30 Draw & Explain the schematic of reactance modulator. (CO4)
- Q.31 What is the need of stabilization of carrier in F.M.? (CO4)
- Q.32 Draw & Explain in brief the diode detector circuit for demodulation of A.M. (CO5)
- Q.33 Show how slope detector works for F.M. detection? (CO6)
- Q.34 State & Explain sampling theorem. (CO7)
- Q.35 Compare T.D.M. with F.D.M. in tabular form. (CO7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Define amplitude modulation, derive expression for amplitude modulated wave. Draw wave forms also. (2+5+3)(CO2)
- Q.37 Draw and explain the balanced modulator circuit. (3+7)(CO4)
- Q.38 With the help of circuit diagram, explain foster-seeley discriminator. (3+7)(CO6)

Note : Course Outcome (CO) mentioned in the question paper is for official purpose only.

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**3rd Semester/
Eltx. IC, Power Eltx., Elect. & Eltx. Engg.
Subject: Principles of Communication Engineering**

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Number of sidebands in A.M. are _____. (CO2)
a) 1 b) 2
c) 3 d) 4
- Q.2 With modulation height of antenna _____. (CO1)
a) Increases b) Decreases
c) Does not change d) None
- Q.3 In F.M. Modulation index is _____ to modulating frequency. (CO3)
a) Equal b) Directly Proportional
c) Inversely proportional d) None
- Q.4 The Maximum frequency deviation in F.M. is _____. (CO3)
a) 10 KHz b) 50 KHz
c) 100 KHz d) 200 KHz

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- Q.5 The balanced modulator produces _____. (CO4)
 a) DSB b) DSB-SC
 c) SSB d) SSB-SC
- Q.6 This _____ modulator is example of Indirect F.M. Generation? (CO4)
 a) Reactance b) Varactor diode
 c) Reactance tube d) Armstrong
- Q.7 If receiver frequency is 2000 KHz, If is 50KHz, then local oscillator frequency will be _____ KHZ. (CO5)
 a) 2050 b) 40
 c) 2100 d) 1000
- Q.8 The ability of a communication system to reject unwanted signal depends upon its _____. (CO5)
 a) Sensitivity b) Selectivity
 c) Fidelity d) Detection
- Q.9 Which is not analog _____. (CO7)
 a) PAM b) PWM
 c) PPM d) PCM
- Q.10 In Phase modulation, modulating frequency is independent of _____. (CO3)
 a) Frequency b) Voltage
 c) Frequency deviation d) Current

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 The band width of A.M. signal is _ (CO2)

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- Q.12 Expand the term VSB. (CO2)
- Q.13 Carrier contributes approximately _____ % age of total power in A.M. (CO2)
- Q.14 In F.M. noise is more than A.M. (True/False) (CO3)
- Q.15 What is value of modulation index in wideband F.M.? (CO3)
- Q.16 Draw a PWM Wave. (CO7)
- Q.17 Pre-emphasis is used to boost _____ frequencies. (High/Low) (CO3)
- Q.18 Collector modulators are used for low power requirements. (True/False) (CO3)
- Q.19 Write the name of any two FM demodulators. (CO5)
- Q.20 Expand AFC. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Draw & Explain a basic communication system. (CO1)
- Q.22 Write a note on VSB. (CO1)
- Q.23 Explain the power distribution concept in A.M. wave. (CO2)
- Q.24 Define DSB-SC, Write its applications & advantages. (CO2)
- Q.25 What are merits of F.M. over A.M.? (CO3)
- Q.26 Explain in brief the working of collector modulator. (CO4)

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